Chapter 16: Genetics and the Brain — Mathematical Equations

Neuroscience Reference

Abstract

This chapter provides mathematical models of genetic inheritance, heritability estimation, polygenic risk architecture, population genetics, linkage disequilibrium, pharmacogenomics, and gene-environment interactions. Key frameworks include Falconer's heritability formula, broad-sense vs. narrow-sense heritability decomposition, Hardy-Weinberg equilibrium, genome-wide association study (GWAS) significance thresholds, polygenic risk scores, DNA methylation kinetics, cytochrome P450 drug metabolism, and liability-threshold models for complex diseases. Detailed case studies quantify twin-based heritability, polygenic score calculation for depression risk, linkage disequilibrium decay, epigenetic methylation dynamics, CYP2C19 polymorphism effects on SSRI dosing, and gene-environment interactions in schizophrenia risk with cannabis use. Mathematical models illustrate that genetic effects are typically small (most variants: OR ~ 1.05), operate through polygenic mechanisms (thousands of variants contributing), and interact non-additively with environmental factors in determining phenotypic outcomes.

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1. Heritability and Twin Studies

1.1 Falconer's Heritability Formula

The most widely used estimate of heritability from twin studies:

$$h^2 = 2(r_{MZ} - r_{DZ})$$

where:

- r_{MZ} = correlation of trait in monozygotic (identical) twins
- r_{DZ} = correlation of trait in dizygotic (fraternal) twins
- h^2 = broad-sense heritability (proportion of phenotypic variance due to genetic variance)

Derivation: MZ twins share ∼100% of DNA; DZ twins share ∼50% on average. Assuming equal environments for both types, the difference in correlation is attributable to genetic factors.

Example: For IQ, typical $r_{MZ} \approx 0.86$ and $r_{DZ} \approx 0.60$, yielding $h^2 = 2(0.86 - 0.60) = 0.52$ (52% heritability).

1.2 Broad-Sense vs. Narrow-Sense Heritability

Broad-sense heritability:

$$H^2 = \frac{V_G}{V_P} = \frac{V_G}{V_G + V_E}$$

Narrow-sense heritability (additive genetic effects only):

$$h^2 = \frac{V_A}{V_P}$$

where:

- V_G = total genetic variance (includes dominance, epistasis)
- V_A = additive genetic variance
- V_E = environmental variance (shared + unique)
- V_P = total phenotypic variance

1.3 Shared and Unique Environment Decomposition

Extended twin model partitioning environmental variance:

$$V_P = V_A + V_D + V_E^{\text{shared}} + V_E^{\text{unique}}$$

Estimating shared environment:

$$c^2 = 2(r_{DZ} - r_{MZ})$$

where c^2 is the proportion of variance explained by shared environment.

Unique environment:

$$e^2 = 1 - h^2 - c^2$$

2. Polygenic Architecture and GWAS

2.1 Polygenic Risk Score (PRS)

Sum of allelic effects across many genome-wide significant loci:

$$\text{PRS} = \sum_{i=1}^{n} \beta_i \cdot A_i$$

where:

- β_i = effect size (log odds ratio or standardized beta) at SNP i from GWAS
- A_i = number of risk alleles (0, 1, or 2) at locus i
- n = number of SNPs included in the score

Clinical application: Individuals in the top 10% of PRS for schizophrenia have ∼3–4× increased lifetime risk vs. population mean. However, PRS remains insufficiently predictive for individual clinical use.

2.2 Variance Explained by Detected SNPs

Proportion of phenotypic variance explained by GWAS-identified variants:

$$R^2 = 2\sum_{i=1}^{n} \beta_i^2 \cdot p_i(1 - p_i)$$

where:

- β_i = standardized regression coefficient at SNP i
- p_i = frequency of the effect allele at locus i

Interpretation: For schizophrenia, ∼300 genome-wide significant loci explain ∼7% of variance on the liability scale. For educational attainment, >4,000 loci explain ∼15% of variance. The "missing heritability" resides in rare variants, gene-gene interactions, and gene-environment interactions.

2.3 SNP Heritability Estimation (GCTA/LDSC)

Genome-wide Complex Trait Analysis estimates heritability explained by common SNPs:

$$h^2_{\text{SNP}} = \frac{\text{Var(PRS)}}{V_P}$$

where Var(PRS) is the variance in polygenic risk scores across the population.

Linkage Disequilibrium Score Regression (LDSC) accounts for LD structure:

$$\chi^2_j = \sum_k r^2_{jk} \cdot h^2 \cdot N$$

where:

- χ^2_j = chi-square statistic for SNP j
- r^2_{jk} = LD between SNPs j and k
- N = sample size
- h^2 = heritability (estimated from slope)

3. Population Genetics (Hardy-Weinberg Equilibrium)

3.1 Hardy-Weinberg Equilibrium

At genetic equilibrium with random mating and no selection:

$$p^2 + 2pq + q^2 = 1$$

where:

- p = frequency of allele A
- q = frequency of allele a (with $p + q = 1$)
- p^2 = frequency of genotype AA
- $2pq$ = frequency of heterozygote Aa
- q^2 = frequency of homozygote aa

Assumptions: Large population, random mating, no mutation, no selection, no migration.

3.2 Allele Frequency Change Under Selection

Change in allele frequency with directional selection:

$$p(t) = \frac{p(0)}{p(0) + q(0) \cdot e^{-st}}$$

or for weak selection:

$$\Delta p = \frac{s \cdot p \cdot q \cdot (p - q)}{1 - 2pqs}$$

where:

- s = selection coefficient (0 = no selection; 1 = lethal)
- t = generations
- $p(0)$ = initial allele frequency

4. Linkage Disequilibrium and Association

4.1 Linkage Disequilibrium (LD) r^2 Measure

Standardized measure of non-random association between two SNPs:

$$r^2 = \frac{D^2}{p_A(1 - p_A) \cdot p_B(1 - p_B)}$$

where:

- $D = p_{AB} - p_A \cdot p_B$ (deviation from random association)
- p_A, p_B = allele frequencies of SNPs A and B
- p_{AB} = frequency of haplotype carrying both A and B alleles
- r^2 ranges from 0 (no LD) to 1 (complete LD)

Interpretation: $r^2 = 0.8$ indicates that 80% of information at locus B is captured by locus A.

4.2 LD Decay with Genomic Distance

Exponential decay of LD as a function of physical distance:

$$r^2(d) = r^2(0) \cdot e^{-\alpha d}$$

or simplified inverse relationship:

$$r^2(d) \approx \frac{0.1}{d/1000 \text{ bp}}$$

where:

- d = physical distance between SNPs (in base pairs)
- α = decay rate parameter (∼0.001–0.01 per kb depending on recombination rate)

Empirical observation: LD halves approximately every 1–2 kb in Europeans (varies by ancestry).

5. GWAS Effect Sizes and Significance

5.1 Log Odds Ratio from Beta Coefficients

Conversion of linear regression coefficient to odds ratio for disease association:

$$\log(\mathrm{OR}) = \beta \cdot 2 \cdot p(1-p)$$

where:

- β = standardized regression coefficient (case-control logistic regression)
- p = effect allele frequency
- OR = odds ratio per copy of effect allele

Alternative: Direct logistic regression yields β as the log-odds change per allele; $\mathrm{OR} = e^{\beta}$.

5.2 Genome-Wide Significance Threshold

Standard multiple-testing correction for GWAS:

$$-\log_{10}(p_{\text{threshold}}) = 7.3$$

or equivalently:

$$p_{\text{threshold}} = 5 \times 10^{-8}$$

Derivation: Bonferroni correction across ∼1 million independent tests; $0.05/1,000,000 = 5 \times 10^{-8}$.

6. Epigenetic Methylation Dynamics

6.1 DNA Methylation Level Changes During Learning and Stress

Exponential approach to equilibrium methylation state:

$$m(t) = m_{\text{eq}} + (m_0 - m_{\text{eq}}) \cdot e^{-kt}$$

where:

- $m(t)$ = methylation level (proportion of sites methylated) at time t
- m_{eq} = equilibrium methylation level (final state)
- m_0 = initial methylation level
- k = rate constant for methylation/demethylation (typically 0.01–0.1 per hour)
- t = time (hours)

Context: Learning activates immediate methylation changes at BDNF, reelin, and PP1 promoters within 1–4 hours. Stress (early adversity) increases glucocorticoid receptor (Nr3c1) promoter methylation, reducing gene expression and impairing stress resilience.

6.2 Histone Acetylation and Gene Expression

Hill-type relationship between histone acetylation and transcriptional activation:

$$\text{Expression} = E_{\max} \cdot \frac{[\text{Ac-Histone}]^n}{K_d^n + [\text{Ac-Histone}]^n}$$

or linearized as:

$$\Delta\text{Expression} = g \cdot \log\left(\frac{[\text{Ac-H3}]}{[\text{H3}]}\right)$$

where:

- [Ac-Histone] = concentration of acetylated histone (proportion of total histone)
- K_d = half-maximal activation histone acetylation level
- n = cooperativity (typically 1–2 for histone acetylation)
- g = slope (effect of acetylation on expression in log scale)

7. Pharmacogenomics and CYP450 Drug Metabolism

7.1 CYP450-Mediated Drug Clearance

First-order pharmacokinetics with CYP-dependent clearance:

$$\frac{dC}{dt} = -\frac{CL}{V_d} \cdot C(t)$$

Solution (single dose):

$$C(t) = \frac{D}{V_d} \cdot e^{-\frac{CL}{V_d} \cdot t} = C_0 \cdot e^{-k \cdot t}$$

where:

- $C(t)$ = plasma drug concentration at time t
- CL = clearance (ml/min/kg) — depends on CYP450 activity
- V_d = volume of distribution
- k = elimination rate constant ($k = CL/V_d$)
- $t_{1/2} = \frac{\ln(2)}{k} = 0.693/k$ (half-life)

7.2 Phenotype-Dependent Steady-State Concentrations

At steady state with repeated dosing:

$$C_{\text{ss}} = \frac{F \cdot D}{CL \cdot \tau}$$

where:

- C_{ss} = steady-state concentration
- F = bioavailability
- D = dose
- CL = clearance (metabolizer phenotype–dependent)
- τ = dosing interval

For poor metabolizers with $CL_{\text{PM}} = 0.25 \cdot CL_{\text{normal}}$:

$$C_{\text{ss,PM}} = 4 \times C_{\text{ss,normal}}$$

8. Gene-Environment Interaction (GxE)

8.1 GxE Variance Decomposition

Phenotypic variance partitioning in presence of gene-environment interaction:

$$V_P = V_G + V_E + V_{GxE} + 2\text{Cov}(G, E)$$

where:

- V_G = genetic variance (independent of environment)
- V_E = environmental variance (independent of genes)
- V_{GxE} = variance due to genotype-by-environment interaction
- $\text{Cov}(G, E)$ = covariance between genetic and environmental factors

Simplified (assuming $\text{Cov}(G, E) \approx 0$):

$$V_P = V_G + V_E + V_{GxE}$$

$$h^2 = \frac{V_G}{V_P}, \quad e^2 = \frac{V_E}{V_P}, \quad (g \times e)^2 = \frac{V_{GxE}}{V_P}$$

8.2 Differential Susceptibility Model

Vulnerability/benefit allocation based on genetic sensitivity to environment:

$$\text{Outcome} = \mu + \beta_G \cdot G + \beta_E \cdot E + \beta_{GxE} \cdot G \cdot E$$

where:

- μ = population mean outcome
- β_G = main effect of genotype
- β_E = main effect of environment
- β_{GxE} = interaction (same sign suggests "orchid" susceptibility; opposite signs suggest diathesis-stress)

Orchid vs. Dandelion interpretation:

- **Orchid genotype**: Performs worse in adverse environments but *better* in supportive environments (synergistic interaction).
- **Dandelion genotype**: Resilient across environments; less affected by adversity.

9. Genetic Risk, Liability-Threshold, and CNV Dosage

9.1 Liability-Threshold Model for Complex Diseases

Disease risk modeled as crossing a threshold on a latent liability distribution:

$$\text{Disease Risk} = P(L > \theta)$$

where:

- L = latent liability (normally distributed)
- θ = disease threshold
- $L = \mu_G + \mu_E + \epsilon$ (genetic + environmental + residual components)

For a trait with heritability h^2 and population prevalence K:

$$\text{Lifetime risk in relative} = \Phi^{-1}(\sqrt{2 \cdot h^2} \cdot \Phi^{-1}(1 - K) + \Phi^{-1}(1 - K_r))$$

where:

- Φ = cumulative normal distribution
- K = population prevalence
- K_r = relative's expected prevalence based on genetic relationship

9.2 Copy Number Variant Dosage Effects

Copy number variants (CNVs) increase risk through copy-number dosage imbalance:

$$\text{Risk} = r_0^{n_{\text{copies}}} \cdot (\text{baseline risk})$$

where:

- n_{copies} = number of copies of risk CNV (normal: 2; deletion: 1; duplication: 3)
- r_0 = risk increase per copy

10. Case Studies

10.1 Case Study 1: Twin-Based Heritability for Personality

Scenario: A twin study of neuroticism reports $r_{MZ} = 0.48$ and $r_{DZ} = 0.24$.

Solution:

$$h^2 = 2(r_{MZ} - r_{DZ}) = 2(0.48 - 0.24) = 0.48$$

Approximately **48% of neuroticism variance is due to genetic differences**.

Shared environment: $c^2 = 2(r_{DZ} - r_{MZ}) = 2(0.24 - 0.48) = -0.48$ (negative; no shared environment effect).

Unique environment: $e^2 = 1 - 0.48 - 0 = 0.52$.

Interpretation: About half of individual differences in neuroticism are genetic; the other half reflects unique experiences. Shared family environment contributes minimally.

10.2 Case Study 2: Polygenic Score Calculation

Scenario: A GWAS of major depressive disorder identified 10 SNPs with effects and an individual's genotypes.

Solution: Calculate PRS:

$$\text{PRS} = \sum_{i=1}^{10} \beta_i \cdot A_i = 0.05 + 0.16 + 0.03 + 0.12 + 0 + 0.07 + 0.02 + 0.18 + 0 + 0.04 = 0.67$$

Total log odds: $\log(\text{OR}) = 0.67$; Odds ratio: $\text{OR} = e^{0.67} = 1.95$.

Interpretation: This individual has $\sim$2$\times$ elevated odds of developing MDD compared to population mean. If population prevalence is 15%, this individual's lifetime risk would be approximately 25–30%.

10.3 Case Study 3: Linkage Disequilibrium Decay

Scenario: Two SNPs separated by 5 kb; SNP-A has $p_A = 0.30$, SNP-B has $p_B = 0.25$, haplotype frequency $p_{AB} = 0.10$.

Solution:

$$D = p_{AB} - p_A \cdot p_B = 0.10 - 0.075 = 0.025$$

$$r^2 = \frac{(0.025)^2}{0.30 \times 0.70 \times 0.25 \times 0.75} = \frac{0.000625}{0.039375} = 0.016$$

Interpretation: $r^2 = 0.016$ indicates essentially no LD. Both SNPs should be included independently.

10.4 Case Study 4: DNA Methylation Dynamics

Scenario: BDNF promoter demethylation post-learning; $m_0 = 55\%$, $m_{\text{eq}} = 35\%$, $k = 0.15$ per hour.

Solution:

$$m(2) = 35 + (55 - 35) \cdot e^{-0.15 \times 2} = 35 + 20 \times 0.741 = 49.82\%$$

$$m(6) = 35 + (55 - 35) \cdot e^{-0.15 \times 6} = 35 + 20 \times 0.407 = 43.14\%$$

Interpretation: Within 2 hours, methylation drops 5%; by 6 hours, 12% demethylation occurs.

10.5 Case Study 5: CYP2C19 and SSRI Dosing

Scenario: Patient is a CYP2C19 poor metabolizer (25% activity) on escitalopram 20 mg daily.

Solution:

For EM at 20 mg: $C_{\text{ss,EM}} \approx 50$ ng/ml

For PM with $CL_{\text{PM}} = 0.25 \times CL_{\text{EM}}$: $C_{\text{ss,PM}} = 50/0.25 = 200$ ng/ml (exceeds toxicity threshold >100 ng/ml).

Adjusted dose: $D_{\text{PM}} = 20 \text{ mg} \times 0.25 = 5$ mg daily.

Interpretation: A CYP2C19 PM should receive 5 mg daily (vs. 20 mg) to achieve therapeutic concentrations without toxicity.

10.6 Case Study 6: GxE in Schizophrenia

Scenario: Cannabis use + COMT genotype predicts schizophrenia risk. Met/Met: 1.2% baseline, 15% with cannabis.

Solution:

Interaction ratio: $\frac{\text{Observed}}{\text{Additive}} = \frac{2.71}{0.78} \approx 3.5$

Interpretation: Met/Met genotype shows 3.5× amplified risk with cannabis use — a diathesis-stress GxE where genetic vulnerability becomes pathological only with environmental exposure.

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